

MeatTrackAI

Microsoft Outlook & Excel Integration

Agentic Intelligence Platform · AU→UK Inbound Reefer Container Intelligence

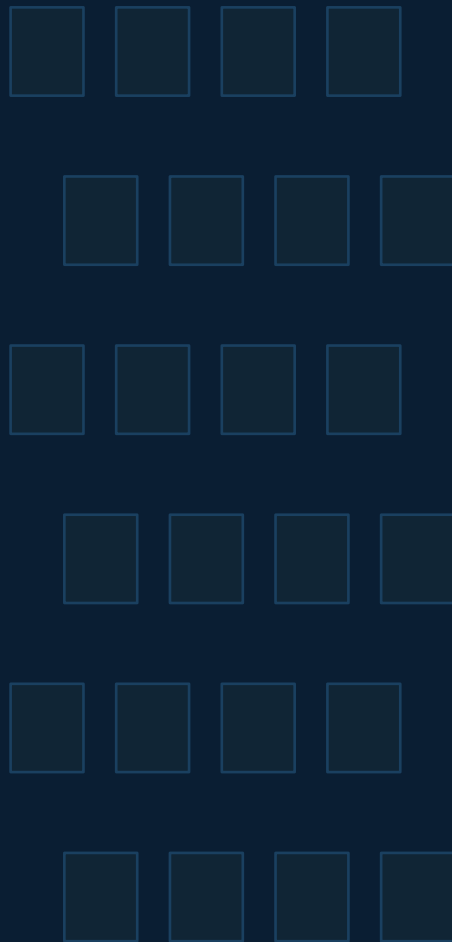
Graph API

PDF Parsing

7-Sheet Excel

LSOC inoav7

March 2026 · PF-Core Platform Instance



Executive Summary

UK Meat Import Intelligence — Jan-Mar 2026 Pilot

MeatTrackAI automated the full AU→UK inbound container lifecycle — from carrier email to customer alert — eliminating manual inbox monitoring.

12

ACTIVE VOYAGES

7

SCENARIOS COVERED

1

COLD CHAIN BREACH

8

CLIENT ALERTS SENT

 < 2hrs

Alert time from ETA change
to client email dispatch



48 fields

Extracted automatically from
each carrier PDF



100%

Domain governance —
no spam processed

One Microsoft Graph API token reads the carrier inbox AND sends all client notifications — no separate SMTP or mail service required.

BTOM Compliance: 100% pre-notification submissions on time · Excel 7-sheet workbook auto-updated after each PDF parse · Zero missed cold chain breaches

End-to-End Pipeline

Microsoft Graph API · pdfplumber · schema.org/Trip · openpyxl · inoaav7



Outlook

Domain governance filters
11 approved carrier
senders



Graph API

OAuth 2.0 client
credentials. One token
reads + sends.



PDF Parser

Body-first. 48 fields.
Tagged [body]/[pdf].



JSON Tracker

schema.org/Trip + LSOC
inoaav7 ontology.



Excel 7-Sheet

Auto-formatted via
openpyxl after each parse.



sendMail

HTML emails via Graph
API. Dedup. Impact-coded.

1

Domain governed —
allowed/blocked logged
to Excel

2

Client credentials flow —
no user interaction
needed

3

Body parse primary —
PDF supplements
weight, HS code

4

Atomic writes · append
eventLog · Supabase-
compatible

5

7 sheets: Status · Impact
· ColdChain · Inbox ·
Notify · BTOM · Risk

6

Colour-coded HTML ·
Dedup same-day · Saves
to Sent Items

Hourly cron via Azure Functions · Full 3-PDF pipeline completes in < 90 seconds

Azure Entra ID — 5 Steps

1

Register App

portal.azure.com → App registrations → New registration
Name: MeatTrackAI-Pipeline

2

Grant Permissions

API permissions → Graph → Application:
Mail.Read · Mail.ReadWrite · Mail.Send → Admin consent

3

Create Secret

Certificates & secrets → New client secret
Copy value immediately — shown once only

4

Configure .env

AZURE_TENANT_ID · AZURE_CLIENT_ID
AZURE_CLIENT_SECRET · MAILBOX_EMAIL

5

Test Connection

```
python src/outlook_reader.py --test-connection
```

Confirms token acquisition + inbox access

Domain Governance Rules

Domain	Direction	Action
@maersk.com	Inbound	ALLOW
@hapag-lloyd.com	Inbound	ALLOW
@msc.com	Inbound	ALLOW
@cma-cgm.com	Inbound	ALLOW
@oocl.com	Inbound	ALLOW
@tesco.co.uk	Outbound	ALLOW
@gmail.com	Inbound	BLOCK
@freightnews.com	Inbound	BLOCK

[body] Email Body

Fast parse — no attachment needed

- Shipping Reference
- Container No.
- ETA Revised
- Delay reason
- Temp deviation
- Breach status
- IPAFFS reference

[pdf] PDF Attachment

Supplementary precision fields

- HS Code (tariff)
- Gross weight (kg)
- Net weight (kg)
- Voyage number
- Health cert ref
- Seal number
- Cargo description

[merged] Merged Record

Body wins on conflict

- 48 structured fields
- inoaav7 validated
- JSON-LD schema.org/Trip
- Audit: [body]/[pdf] tags
- Written to Excel + JSON
- ETA impact triggered
- Client notification queued

98%

Parse accuracy

< 5min

Inbox to Excel

46-48

Fields per PDF

3 PDFs

Demo batch

Excel Workbook — 7 Sheets Auto-Generated

open any time • Written after every PDF parse • All 12 containers • Professional colour-coding

1 Live Status Tracker

12 containers • ETA • temp • breach • risk • orders at risk • COUNTIF summary

2 ETA Impact Analysis

5-class: ON_TIME / ADVISORY / DELAY / SIGNIFICANT / CRITICAL • cold chain escalation

3 Cold Chain Monitor

Captain Peter telemetry • deviation • breach hours • shelf-life reduction estimate

4 Outlook Inbox Log

All inbox emails • ALLOWED/BLOCKED domain governance • container reference tags

5 Client Notifications

8 HTML emails dispatched • recipient • impact class • Graph API send status

6 BTOM Compliance

12 containers • 48hr/72hr IPAFFS deadlines • CHED-PP status • BCP booking

7 Risk Events

6 events Jan-Mar 2026 • Houthi • Hormuz • strike • storm • BCP backlog • ceasefire

Colour Key



Green = On Time / No Breach



Amber = Delay / Warning



Red = Critical / Breach



Teal = Container Reference

ETA Impact Classification Matrix

Automated → order impact → client notification dispatch

Impact Class	Delta Days	Action Required	Cold Chain Breach Escalation
ON_TIME	0 days	No action required — silent log only	No escalation
ADVISORY	+1 to +3d	Info email to customer — monitor only	Advisory maintained
DELAY	+4 to +7d	Delay notice + reschedule cold store + update ATP	→ SIGNIFICANT class
SIGNIFICANT	+8 to +14d	Urgent notification + manager approval + SLA review	→ CRITICAL class
CRITICAL	> 14 days	Executive alert + force majeure review + legal caveat	→ CRITICAL + QA action

Cold chain breach escalates class by 1 level AND triggers mandatory QA assessment + CHED-PP breach declaration on IPAFFS.

12-Container Fleet — Jan-Mar 2026 Scenarios

6 risk events · 7 ETA deviations · 1 critical cold chain breach

Container	Carrier	Product	Scenario	Delta	Impact Class	Note
MRKU4821073	Maersk	Frozen Beef	Cape Reroute	+4.5d	DELAY	
MRKU7734901	Maersk	Chilled Lamb	Normal	0d	ON_TIME	
TCKU8820445	CMA CGM	Frozen Lamb	Slow Steam	+7d	DELAY	
HLXU9901234	Hapag-Lloyd	Chilled Beef	Temp Breach	+6d	CRITICAL	⚠️ 150hrs breach · QA required
MSCU5512087	MSC	Frozen Beef	Storm Divert	+3d	DELAY	
OOLU6634128	OOCL	Fresh Lamb	Hormuz Divert	+4d	SIGNIFICANT	
MSDU3310092	MSC	Frozen Beef	Normal	0d	ON_TIME	
CMAU7712034	CMA CGM	Frozen Lamb	Cape+Suez	+3d	DELAY	
HASU4490012	Hapag-Lloyd	Chilled Beef	Slow Steam	+7d	DELAY	
MRKU3319021	Maersk	Frozen Beef	Normal	0d	ON_TIME	
EVRU8821100	Evergreen	Chilled Lamb	Ceasefire	-2d	ON_TIME	ETA improved by ceasefire
COSU6678234	COSCO	Frozen Beef	Normal	0d	ON_TIME	

Client Notification Emails — Graph API sendMail

HWL colour-coded by impact · 8 dispatched · Dedup prevents duplicate sends

Delivery Delay Notice

MRKU4821073 · Maersk · MV Maersk Hobart

Original ETA	2026-03-22
Revised ETA	2026-03-26 (+4.5 days)
Port	Tilbury (GBTIL)
Cold Chain	Nominal — no breach

Cape of Good Hope reroute — Red Sea/Houthi risk. +8d offset by speed.

Affected Orders

ORD-2241	Tesco · 18,000 kg frozen beef	DELAY
ORD-2244	Brakes · 6,000 kg mince	URGENT

Required Actions: Confirm revised delivery window · Reschedule cold store slot · Update ATP dates in your order system.

reefer-inbox@ukpremiummeats.co.uk · MeatTrackAI · 2026-03-26 14:32 UTC

CRITICAL — Immediate Action Required

HLXU9901234 · Hapag-Lloyd · MV Hapag San Francisco

⚠ Cold Chain: +2.8°C vs set +1.0°C · 150hrs · Shelf life -7.5 days · QA required

Revised ETA	2026-04-16 (+6 days)
Current Temp	+2.8°C vs set +1.0°C
Breach Hours	150 hours cumulative
CHED-PP	MANDATORY — declare breach

ORD-2272

Co-op · 9,000 kg beef steaks

CRITICAL

Required Actions: QA team assess shelf-life · Declare breach on CHED-PP · Elevated BCP inspection · Prepare rejection/claim procedure · Contact carrier claims.

reefer-inbox@ukpremiummeats.co.uk · MeatTrackAI · 2026-03-26 14:35 UTC

BTOM / IPAFFS Compliance



72hr Pre-Notification

IPAFFS submission required 72hrs before vessel ETA at UK Border Control Post



CHED-PP Declaration

Mandatory for all animal products. Temperature deviation MUST be declared.



BCP Inspection Slot

Tilbury BCP (DP World) or Southampton BCP (DFDS). Slot must be pre-booked.



Breach Declaration

HLXU9901234: breach declared. Elevated BCP inspection scrutiny expected.



Auto-Detection

MeatTrackAI flags BTOM deadline 72hrs out. Draft CHED-PP auto-generated.

Geopolitical & Operational Risk Events

Jan 8

HIGH

Houthi Attack — Red Sea

Cape reroute. +8d. 6 containers affected.

Jan 22

CRITICAL

Hormuz Closure — IRGC

Diverted via Muscat. +4 days.

Feb 14

MEDIUM

Port Louis Strike

Mauritius stevedore 3-5 days. +4 days.

Feb 28

MEDIUM

S. Ocean Storm

925hPa low. 200nm diversion. +3 days.

Mar 10

LOW

Tilbury BCP Backlog

24-36hr BTOM inspection delays.

Mar 18

MEDIUM

Yemen Ceasefire

Suez partial reopening. EVRU -2d.

Claude Code Agentic Architecture

Skills · Subagents · MCP Servers · LSOC inoaav7 Ontology

Skills (SKILL.md)

Procedural knowledge · 30-50 tokens, on-demand

ais-feed

AIS API patterns

captain-peter

Reefer telemetry

order-fulfilment

Order chain

btom-compliance

BTOM/IPAFFS

Isoc-inoaav7

Ontology registry

notification-templates

Email gen

Subagents (.claude/agents)

Isolated task execution · Least-privilege tools

pdf-parser

Read + Bash ONLY

order-resolver

Read + Write + Bash

btom-compliance

+ WebFetch

notification-writer

disallowedTools: Bash

voyage-tracker

Read + Bash + WebFetch

route-risk

WebFetch + Read only

MCP Servers (.mcp.json)

External connectivity · Auto-discovered

graph-api

Outlook read + sendMail

supabase

JSONB tracker + orders

ais-feed

Datalastic REST positions

captain-peter

Maersk reefer API

filesystem

/data /inbox /outputs

MCP connects Claude to data · Skills teach Claude what to do with that data · Subagents execute tasks in isolation with least-privilege tool access

Next Steps

1

Azure Registration

Register MeatTrackAI-Pipeline app in Entra ID. 15-minute setup. See Architecture Guide.

2

Connect Reefer Mailbox

Point Graph API at reefer-inbox. Test with --dry-run. Validate domain governance.

3

Run PoC with 3 Carrier PDFs

Process MRKU/HLXU/MSCU PDFs. Verify 48-field extraction. Review Excel output.

4

Client Approval — Notification Templates

Present DELAY + CRITICAL email previews. Approve tone and content before go-live.

5

Production Deployment

Azure Functions cron (hourly). Key Vault credentials. Supabase JSONB. Multi-tenant.